Enrollment No._____

GANPAT UNIVERSITY

B.Sc (CA & IT) Sem-III Reg\Rem Nov-Dec 2024

U13A1DS: DATA STRUCTURE

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Explain the difference between primitive and non-primitive data structures.	06	CO1	BTL-2
2	Explain the concept of "Push" and "Pop" operations in a stack.	06	CO2	BTL-2
3	What are the Differences Between Simple Queue and Circular Queue?	06	CO2	BTL-2
4	Describe the operations of inserting and deleting a node in a singly linked list.	06	CO3	BTL-2
5	Draw a small threaded binary tree with three nodes and show the placement of threads.	06	CO3	BTL-3
6	Evaluate the advantages and disadvantages of using an adjacency list versus an adjacency matrix for graph representation.	06	CO3	BTL-5
7	Explain how Bubble Sort works, using an example with a small set of numbers.	06	CO4	BTL-2
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Convert the infix expression (A + B) * (C - D) to postfix using a stack.	06	CO1	BTL-3
2	Define a circular queue and a priority queue.	06	CO2	BTL-1
3	Evaluate the performance of linked lists in terms of time complexity for different operations like insertion and deletion.	06	CO2	BTL-5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the concept of traversal in trees and list the different traversal methods.	06	CO3	BTL-2
2	what are the different types of graph representations?	06	CO3	BTL-1
3	Which search technique would you use for a large dataset with frequent updates, and why?	06	CO3	BTL-5
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain the process of Selection Sort using an example with an array of 4 elements.	06	CO4	BTL-2
2	Describe the difference between quicksort and merge sort.	06	CO4	BTL-1

END OF PAPER

GANPAT UNIVERSITY

B.Sc. (CA & IT) Sem-III Regular/Remedial Nov-Dec 2024

U13A2OCP: OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	What is polymorphism in OOP? Give an example.	06	CO1	BTL4
2	What is a constructor in Java? How does it differ from a method?	06	CO1	BTL4
3	Explain the concept of exception handling and provide a Java example that demonstrates using try, catch, and finally blocks.	06	CO2	BTL6
4	Explain the purpose of packages in Java and show how to define a package with a simple example.	06	CO3	BTL5
5	Describe the Vector class and provide an example showing how to use it.	06	CO3	BTL6
6	Explain constructor references and demonstrate their use with a simple example.	06	CO4	BTL6
7	Explain the fork/join framework and provide an example demonstrating how it is used for parallel programming.	06	CO4	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Compare the String class and StringBuffer class in Java.	06	CO1	BTL4
2	Explain the concept of inheritance and give a basic Java code snippet to illustrate how one class inherits from another.	06	CO2	BTL5
3	Explain abstract classes and abstract methods, and describe their use with an example.	06	CO2	BTL6
Q.3	**Answer the following: (Any Two)**	**12**		
1	Define generics and explain their purpose. Provide an example of a generic class.	06	CO3	BTL4
2	Explain how lambda expressions can handle exceptions and give an example demonstrating this.	06	CO4	BTL6
3	List and briefly describe predefined functional interfaces in Java.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Describe synchronization objects and provide an example demonstrating their use to ensure thread safety.	06	CO4	BTL4
2	Explain the purpose of Future and Callable interfaces in the Executor framework and how they are used.	06	CO4	BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc (CA & IT) Sem-III Reg\Rem Nov-Dec 2024

U13A3DMS: DATABASE MANAGEMENT SYSTEM

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Explain briefly about three levels Architecture of database system.	06	CO1	BTL3
2	What is ER model? Explain various component of ER model.	06	CO2	BTL3
3	Discuss group by and having clause with example	06	CO4	BTL4
4	Explain any five-Aggregate function with example.	06	CO4	BTL3
5	Write short note on Sequence.	06	CO5	BTL3
6	What is normalization? Explain 1NF with example	06	CO2	BTL4
7	Discuss ALTER command with all options.	06	CO3	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	What is data model? explain Hierarchical Data model.	06	CO1	BTL3
2	Write short note on 5NF with example.	06	CO2	BTL3
3	Explain relation model with example.	06	CO1	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	What is constraint? Explain primary key with example.	06	CO3	BTL3
2	Discuss SET operators with example.	06	CO4	BTL4
3	What is joins? Explain types of joins.	06	CO4	BTL3
Q.4	**Answer the following: (Any One)**	**06**	**CO**	**Bloom's Taxonomy**
1	Explain indexes with example.	06	CO5	BTL3
2	Discuss grant and revoke with example.	06	CO5	BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc (CA & IT) Sem-III Reg\Rem Nov-Dec 2024

U13A4SAD SYSTEM ANALYSIS AND DESIGN

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

	Section – I			
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Explain the difference between System Analysis and System Design.	06	CO1	BTL-2
2	Describe the SSADM methodology and its significance in system design.	06	CO2	BTL-2
3	Describe the differences between Physical DFD and Logical DFD.	06	CO3	BTL-2
4	What are the challenges involved in code design when dealing with large data systems?	06	CO3	BTL-4
5	Compare white box and black box testing in terms of advantages and when each is suitable.	06	CO4	BTL-4
6	Assess the importance of documentation throughout the testing process and system implementation.	06	CO4	BTL-5
7	What is a College Management System? List its main features	06	CO5	BLT-1
	Section – II			
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Describe the role of a System Analyst in system development.	06	CO1	BTL-1
2	Identify and explain the elements of a system.	06	CO2	BTL-1
3	Identify the symbols used in a Data Flow Diagram (DFD) and explain their meanings.	06	CO3	BTL-1
Q.3	**Answer the following: (Any Two)**	**12**		
1	Why is code design important in a system?	06	CO3	BTL-2
2	Design a sample form layout for a customer feedback form.	06	CO3	BTL-3
3	Why is documentation essential in system implementation?	06	CO4	BTL-2
Q.4	**Answer the following: (Any One)**	**06**		
1	What are the typical requirements of an Online Examination System?	06	CO5	BTL-1
2	Explain how a Hotel Management System might handle reservations and billing processes.	06	CO5	BTL-2

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc (CA & IT) Sem-III Reg\Rem Nov-Dec 2024

U13A5OR Operation Research

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy
1	Use the graphical method to solve the following Linear Programming Problem. $$Max\ Z = 15x_1 + 10x_2$$ Subject to constraints, 1. $x_1 + 2x_2 \leq 10$ 2. $x_1 + x_2 \leq 6$ 3. $x_1 - x_2 \leq 2$ 4. $x_1 - 2x_2 \leq 1$ and $x_1, x_2 \geq 0$	06	CO1	BTL3
2	Solve following Linear Programming Problem by simplex method. $$Max\ Z = 4x_1 + 3x_2$$ Subject to constraints, 1. $2x_1 + x_2 \leq 1000$ 2. $x_1 + x_2 \leq 800$ 3. $x_1 \leq 400$ 4. $x_2 \leq 700$ and $x_1, x_2 \geq 0$	06	CO1	BTL5
3	Apply MODI method to obtain optimal solution of transportation problem using following data.	06	CO3	BTL3

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	34

4 Find initial basic feasible solution using North Weast Corner Method. 06 CO2 BTL5

Distribution centers

PLANTS	D_1	D_2	D_3	D_4	Supply
P_1	2	3	11	7	6
P_2	1	0	6	1	1
P_3	5	8	15	9	10
Demand	7	5	3	2	

5 Find initial basic feasible solution using Vogel's Approximation Method. 06 CO2 BTL3

Destinations

ORIGIN	D_1	D_2	D_3	D_4	Supply
O_1	11	13	17	14	250
O_2	16	18	14	10	300
O_3	21	24	13	10	400
Demand	200	225	275	250	

6 A department of a company has five jobs to performed. The time (in hours) that each man takes to perform each job is given in the effective matrix. How the job should be allocated each man to minimize total time. 06 CO4 BTL4

JOB

MAN	I	II	III	IV	V
A	10	5	13	15	16
B	3	9	8	13	6
C	10	7	2	2	2
D	7	11	9	7	12
E	7	9	10	4	12

7 A Book binder has one printing press, one binding machine and manuscripts of 7 different books the time required performing printing and binding operation for different books are shown below. Find total elapsed time. Also find Idle time for printing and binding machines. 06 CO5 BTL5

Book	1	2	3	4	5	6	7
Printing time (hours)	20	90	80	20	120	15	65
Binding time(hours)	25	60	75	30	90	35	50

Section – II

Q-2 Answer the following (Any Two)

		Marks	CO	Bloom's Taxonomy
		12		
1	Describe History of Operation Research.	06	CO1	BTL1
2	Use the graphical method to solve the following Linear Programming Problem.	06	CO1	BTL3

$$Max\ Z = 40x_1 + 80x_2$$

Subject to constraints,

1. $2x_1 + 3x_2 \leq 48$
2. $x_1 \leq 15$
3. $x_2 \leq 10$ and $x_1, x_2 \geq 0$

3 Find initial basic feasible solution using Least cost Method. 06 CO2 BTL3

Distribution centers

PLANTS	D_1	D_2	D_3	D_4	Supply
P_1	2	3	11	7	6
P_2	1	0	6	1	1
P_3	5	8	15	9	10
Demand	7	5	3	2	

Q.3 Answer the following: (Any Two) **12**

1 A factory has five work to performed. How the machine should be allocated each person to minimize total time. Time(in hours) is given in the effective matrix. 06 CO4 BTL5

MACHINE

PERSON	I	II	III	IV
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

2 Apply MODI method to obtain optimal solution of transportation problem using following data. 06 CO3 BTL3

	A	B	C	Supply
W	4	8	8	76
X	16	24	16	82
Y	8	16	24	77
Demand	72	102	41	

3 Solve the following Assignment Problem and find maximum profit. 06 CO4 BTL5

SALES

SALESMAN	I	II	III	IV
A	42	35	28	21
B	30	25	20	15
C	30	25	20	15
D	24	20	16	12

Q.4 Answer the following: (Any One) **06**

1 Find an optimal sequence for the following sequencing problems for four jobs and five machines. When passing is not allowed its processing time(in hours) given below. 06 CO5 BTL6

Job	A	B	C	D
Machine M_1	7	6	5	8
Machine M_2	5	6	4	3
Machine M_3	2	4	5	3
Machine M_4	3	5	6	2
Machine M_5	9	10	8	6

2 We have five jobs, each of which must be processed on the two machines A and B in the order AB. Processing times in hours are given in the table below. Determine a sequence for the five jobs that will minimize the elapsed time T. 06 CO5 BTL6

Job	1	2	3	4	5
Machine A	5	1	9	3	10
Machine B	2	6	7	8	4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DATA SCIENCE)- III Reg\ Rem – Nov-Dec. 2024

U73B6EDM: ENVIRONMENT DISASTER MANAGEMENT

Time: 3 Hours **Total: 60 Marks**

Instructions:

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6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	How nature balances in between their elements? Explain.	06	CO1	BTL4
2	What is ecosystem? Explain ecosystem within different spheres of earth.	06	CO2	BTL5
3	Explain the Green House Effect and term Global Warming.	06	CO3	BTL4
4	Discuss disaster management cycle.	06	CO4	BTL5
5	Explain the scope of environment.	06	CO1	BTL3
6	Evaluate the reason for acid rain. How it effects on life?	06	CO3	BTL5
7	Illustrate the planning for Earthquakes and Landslides disasters.	06	CO5	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	What is environment define the term and explain in detail.	06	CO1	BTL4
2	Define the term ecology and explain it.	06	CO2	BTL5
3	Define: Producer, Consumer and Decomposer.	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the common sources of air pollution? How can we control air pollution?	06	CO4	BTL5
2	Explain the effects on human life due to different pollutions.	06	CO3	BTL3
3	Give the classification of Hazards. List and brief five hazards in each of the category.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain disaster due to nuclear power accidents.	06	CO5	BTL3
2	Write short notes on war and Terrorisms.	06	CO5	BTL3